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Where to Find GrALoG & Installation

Where to find GrALoG

The GrALoG software can be found and downloaded at their GitHub page, linked Here.

The instructions regarding the installation of GrALoG are given in their GitHub README.

GrALoG uses Gradle build 4.10.2, and this cannot be overwritten.

In order for GrALoG to run, JavaSE-11 (version 11) must be used.

The GrALoG Software

The Ford-Fulkerson Plugin files

The files relevant to this plugin are available Here, along with instructions on how to incorporate its files into the GrALoG workspace.

Additionally, a precompiled version of GrALoG with the Ford-Fulkerson plugin included is available from the same location.

Using the Plugin: The Ford-Fulkerson Algorithm

The Ford-Fulkerson algorithm is an algorithm which computes the maximum flow from source to sink in a network graph. That is to say, it finds the maximum flow from point A to point B on a directed graph, where point A is a vertex with no incoming edges, point B a vertex with no outgoing edges, and all edge weights from vertex U to vertex V represent the capacity from U to V.

The Ford-Fulkerson Algorithm computes a maximum flow in an iterative manner by analysing the graph at each pass, searching for augmenting paths that increase the flow from its current state.[1] Two requirements must be met for a flow to be considered valid and for an augmenting path to be found: each edge may only carry its maximum capacity, and all nodes must have equal incoming and outgoing flow when summing the flow across a network node.

This plugin utilises a breadth-first search during the path-finding section, ensuring that a maximum solution is found efficiently. This approach to the Ford-Fulkerson method is also known as the Edmonds-Karp algorithm.[x]

1. Ford, L. R. Jr. and D. R. Fulkerson. Flows in Networks. Princeton University Press, 1962.
2. Edmonds, J., Karp, R.M.: Theoretical improvements in algorithmic efficiency for network flow problems. 1972

Using the Plugin: The Ford-Fulkerson Algorithm

The residual graph is a key concept in understanding the algorithm; this graph is constructed through the use of “forward” edges which show the remaining capacity of the edge in the direction of flow from vertex U to vertex V, and “backwards” edges that show how much flow has already been pushed from vertex U to vertex V. This “backwards” edge points in the opposite direction to the direction of flow, so in this example would point from vertex V to vertex U. This plugin handles the residual graph building for you, so only concern yourself with building your desired network graph!

If you were to partition the vertices of a network graph in two, such that all source vertices were on one side of the partition and all sink vertices in the other, the edges from the source partition to the sink partition are referred to as “cut” edges, and the sum of the capacities of these edges is the size of this cut.

Within network graphs, there is a concept known as the “Max-Flow-Min-Cut” theorem. This theorem states that the maximum flow through a network graph is equal to the minimum sized cut of its edges[3] - that is to say, there is a partition wherein the only edges that cross from source side to sink side have 0 remaining capacity, also known as bottleneck edges.

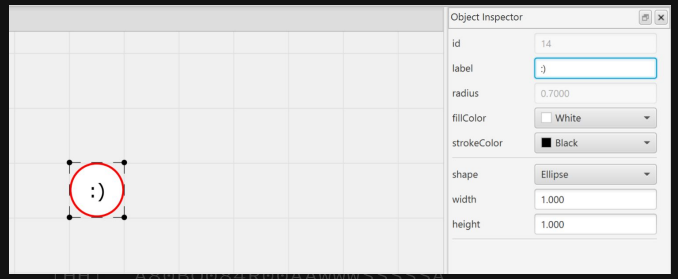
This partition can be identified within the Ford-Fulkerson algorithm by analysing the residual graph once the maximum flow is found. Starting from a source vertex, check all outgoing edges (“forward” or “backward”) for an edge with any value greater than 0. Follow that edge to its destination vertex, and add this vertex to the source side of the partition. Repeat this process for all vertices in the source side of the partition, until there are no more vertices reachable from a source-side vertex by an edge with any value greater than 0. All remaining vertices are then contained in the sink-side partition.

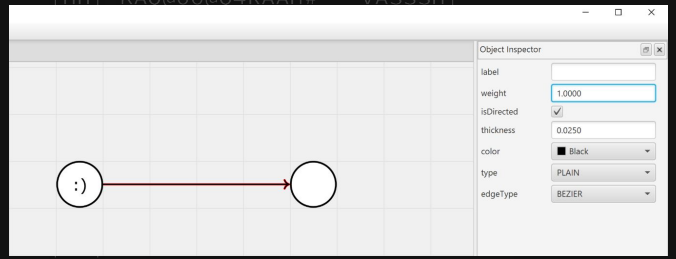
3. J. Yuan, E. Bae and X. -C. Tai, "A study on continuous max-flow and min-cut approaches," 2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition

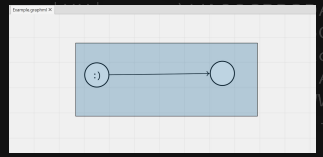
Using the Plugin: Using GrALoG

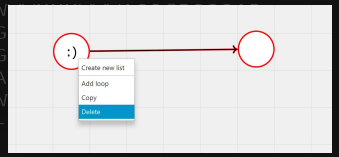
When using GrALoG, vertices can be added by double clicking anywhere on the gridded area. Vertices can be labelled, once selected, by typing in the “Label” textbox in the Object Inspector panel.

Edges can be added by right clicking (and holding) on a vertex, and dragging to your desired destination vertex; loops can be added by right clicking your desired vertex and selecting the “loop” option in the menu which appears.
Edge weights can be changed in the “weight” textbox in the Object Inspector panel - note that there is a separate “label” textbox: this function is utilised by the Ford-Fulkerson plugin, so you do not need to interact with it.
The GrALoG software defaults to a directed graph construction, which is necessary for the use of this plugin.
If you have made a mistake and wish to erase some aspect of your graph, right click on the gridded area and drag over each undesired aspect. Once selected, right click and click “Delete” in the menu which appears.



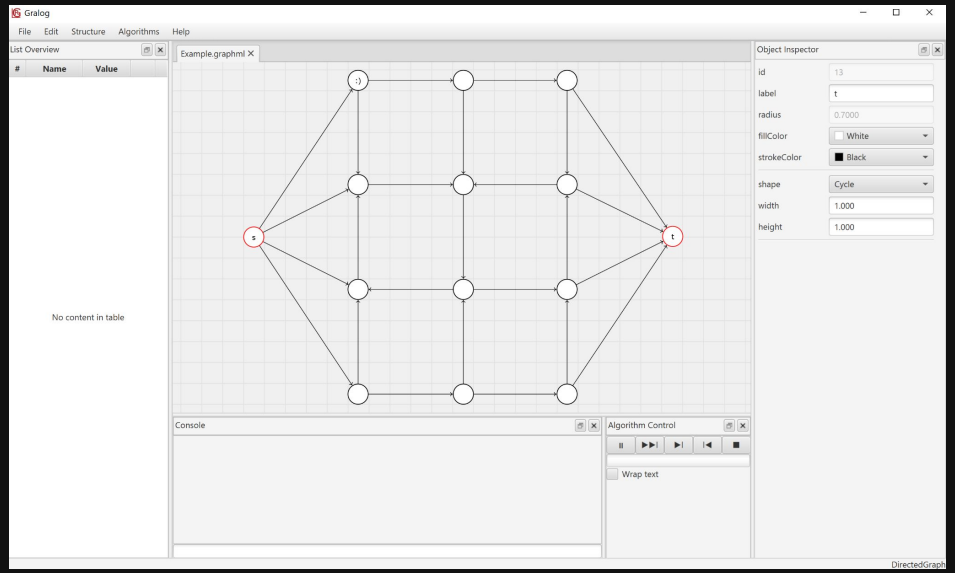


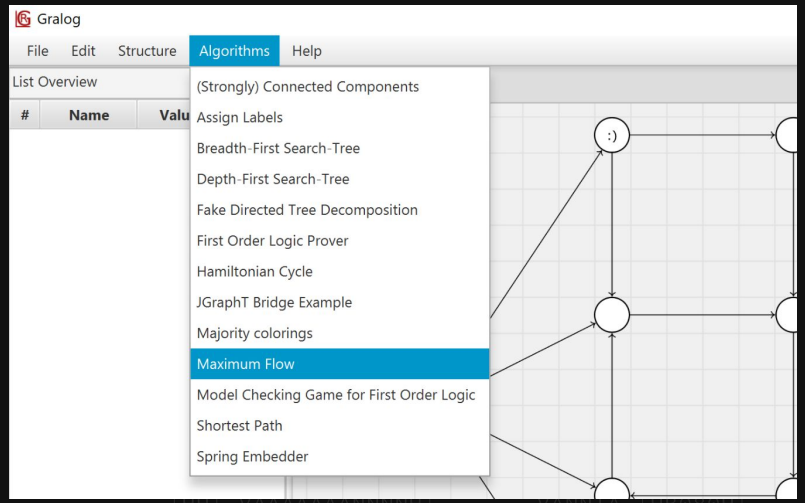




Using the Plugin

Once your graph is drawn, ensure that your sources are labelled with an “s” (and have no incoming edges), and your sinks with a “t” (and have no outgoing edges).

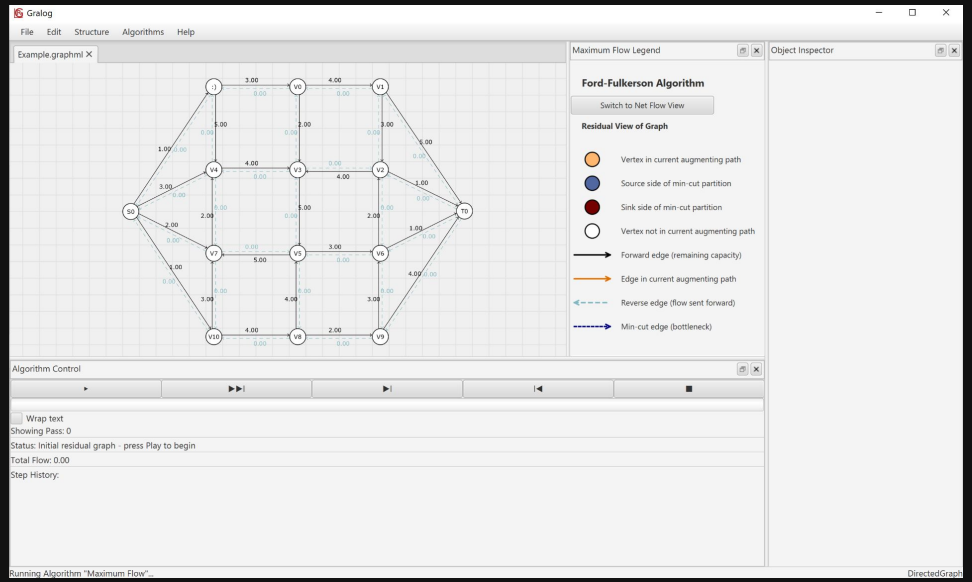


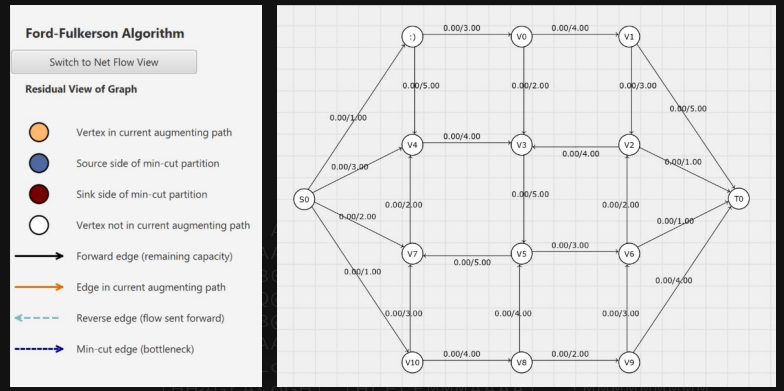


When you are ready to perform the Ford-Fulkerson algorithm on your graph, navigate to the Algorithm drop-down menu and select “Maximum Flow”.

Using the Plugin

After selecting “Maximum Flow”, your graph should be altered to its residual network form. GrALoG should appear something like this:

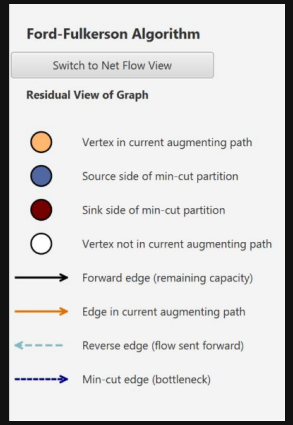


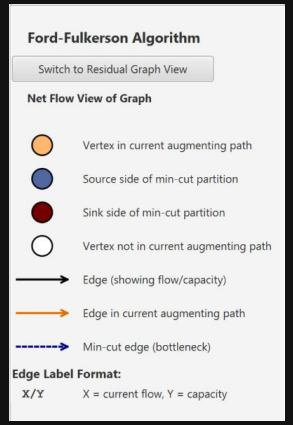


Net flow view, or flow/capacity view.

If you would prefer the net flow view for your visualisation, the button labelled “Switch to Net Flow View” in the legend panel allows for just this. Once selected, the residual graph view can be recovered at any time from the same location.

Using the Plugin





The two different legend versions [left] show the relevant information for each graph view.

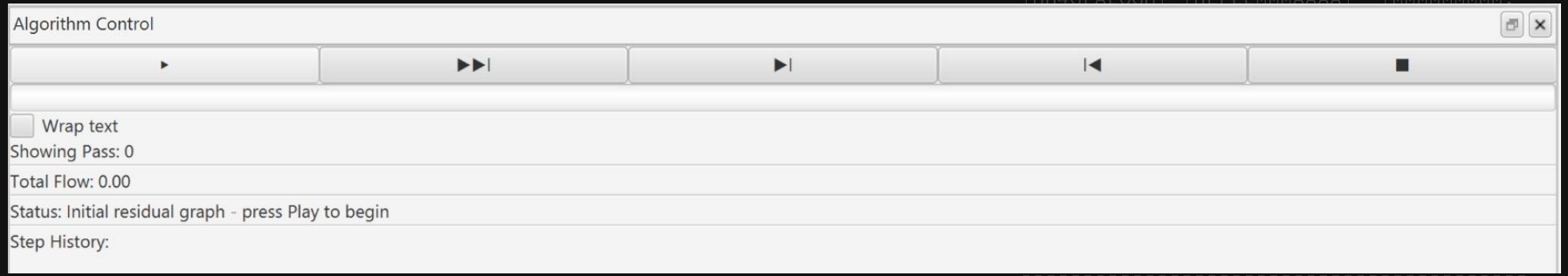
The buttons of the algorithm control panel, control panel depicted below, is used for controlling the pace of the visualiser.

“Showing Pass” communicates which pass of the algorithm is currently being shown.

“Total Flow” gives the total found flow from source(s) to sink(s) at the current pass.

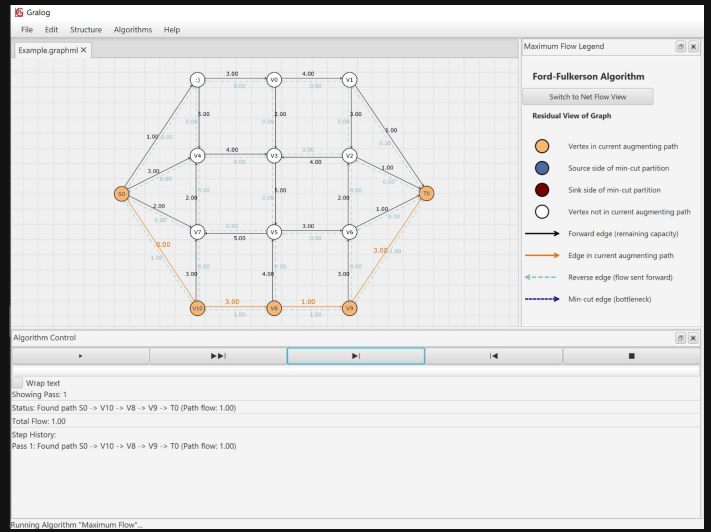
“Status” explicitly communicates the operation performed to reach the shown stage of the graph.

“Step History” contains a running log of each operation from pass 0 to the currently shown pass.



Using the Plugin

Using step forward for first pass on example graph.
Orange edges show edges in current augmenting path.
Dark green vertices show vertices in current augmenting path.



Pressing play (►) sets the algorithm progressing in 2-second increments. This will progress until the final step of the algorithm, which then shows a min-cut partition of your graph.

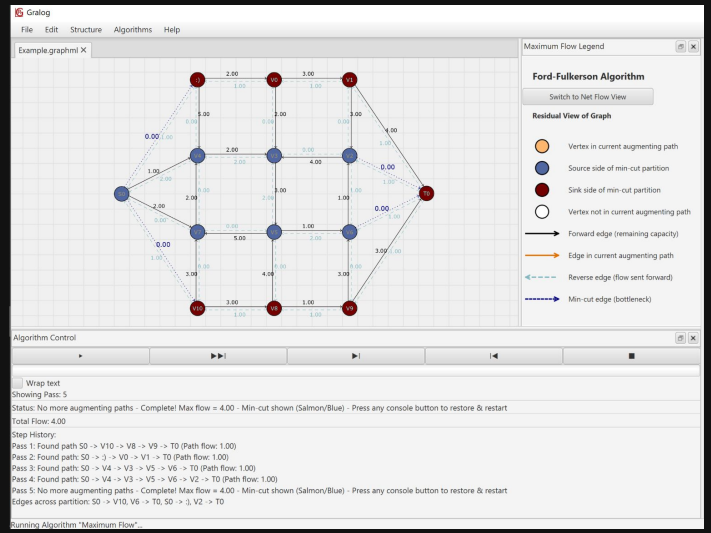
Pausing (❙❙) during play will freeze the algorithm at its current pass until another button is pressed.

Pressing skip to end (►►❙) progresses through all steps of the algorithm, and displays the found min-cut partition of your graph. All passes will be described in the “Step History” segment, and the edges across the cut will be given explicitly.

Pressing step forward (►❙) takes you to the next pass of the algorithm, and pauses.
Selecting step backward(❙◄) takes you to the previous pass of the algorithm, and pauses.
Stopping (■) the algorithm during play will reset the graph to its initial version.
Stopping the algorithm again from its initial state will abort the algorithm and restore your original graph.

Using the Plugin

Using skip to end on the example graph.
Red, dotted edges show the edges across the minimum cut of the graph.



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